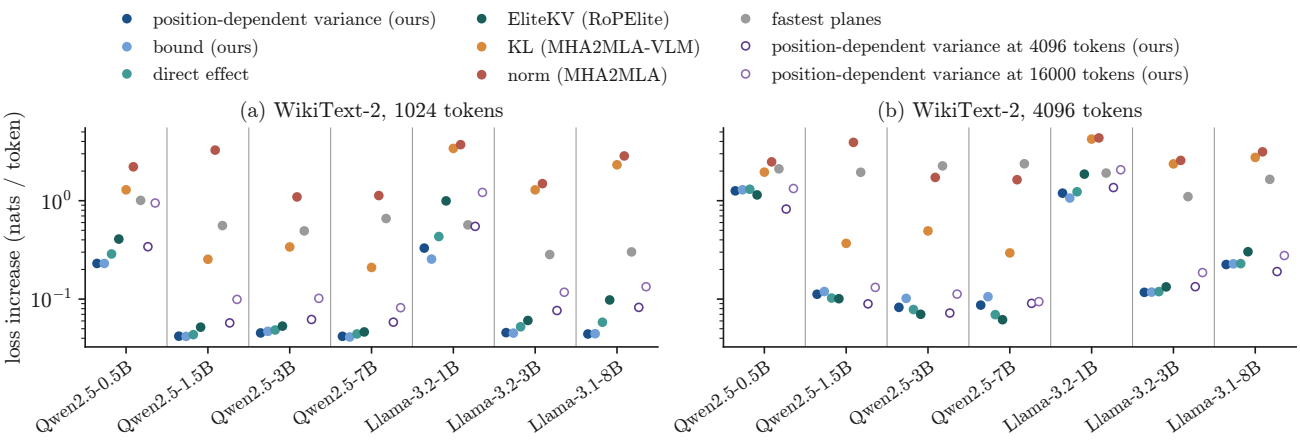
position-dependent variance (ours)
bound (ours)
direct effect
EliteKV (RoPElite)
KL (MHA2MLA-VLM)
norm (MHA2MLA)
fastest planes
position-dependent variance at 4096 tokens (ours)
position-dependent variance at 16000 tokens (ours)
(a) WikiText-2, 1024 tokens
(b) WikiText-2, 4096 tokens
loss increase (nats / token)
10^0
10^{-1}
Qwen2.5-0.5B
Qwen2.5-1.5B
Qwen2.5-3B
Qwen2.5-7B
Llama-3.2-1B
Llama-3.2-3B
Llama-3.1-8B